Hybrid MRI Slice Reconstruction

Combining Classical Interpolation and Deep Learning

A Framework for High-Fidelity 3D Medical Imaging

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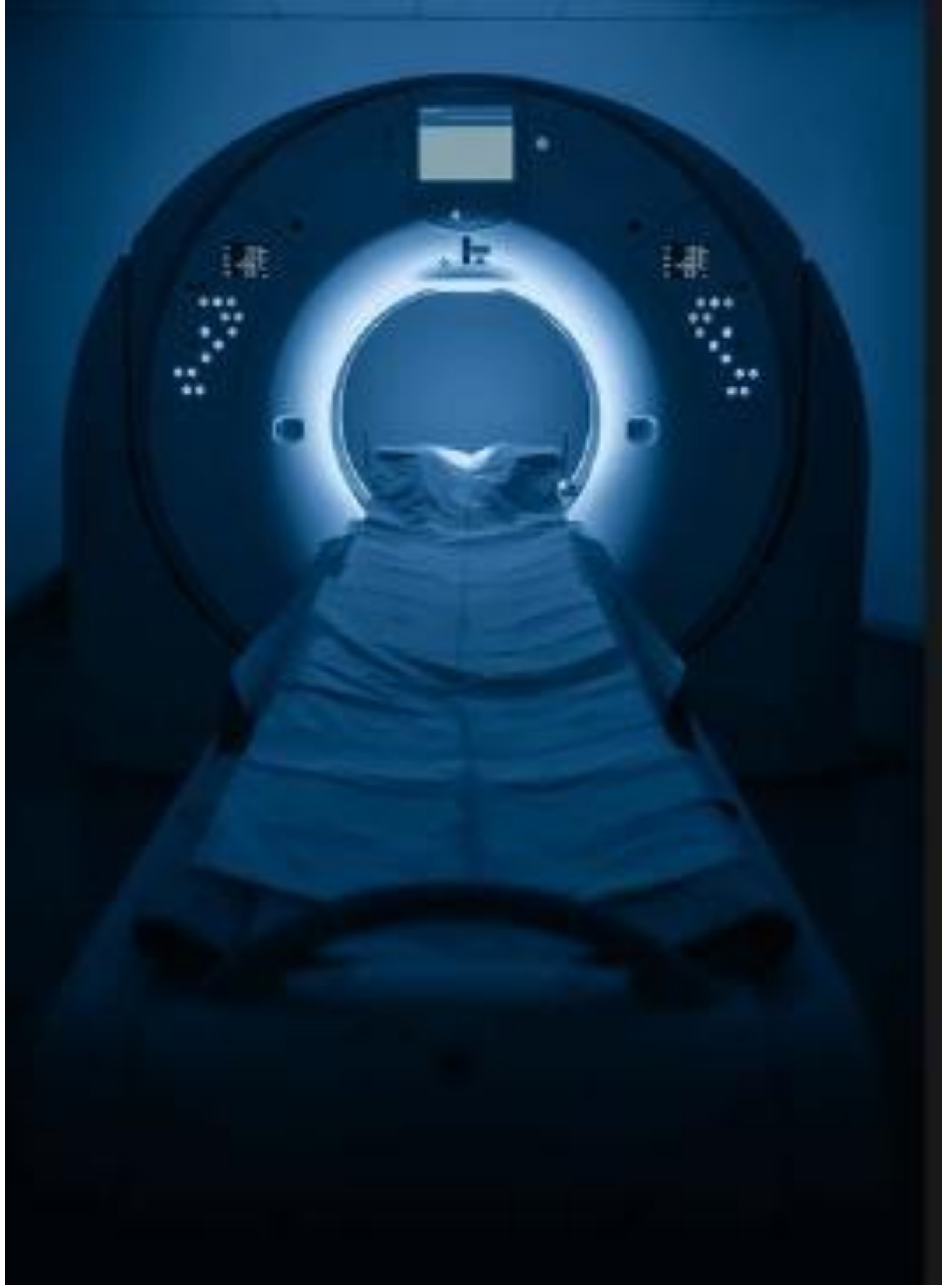


The Missing Data Bottleneck in 3D MRI

Fast Scans = Gaps

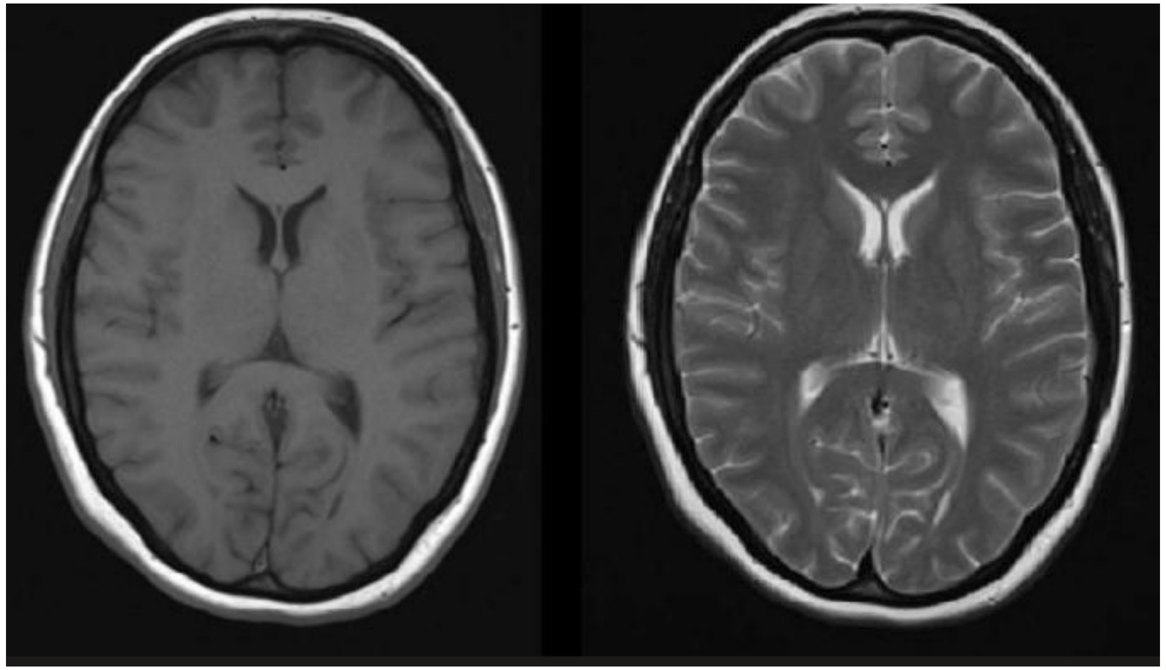
Quick MRI scans save time, but they leave physical "gaps" of missing data between the pictures.

The "Blocky" 3D Effect

While one 2D slice looks perfect, stacking them together creates a jagged, low-quality 3D shape.

The Clinical Impact

This missing information makes it much harder to accurately render the brain or detect tumors.



Limitations of Classical Methods

Linear Interpolation

Speed: Computationally efficient

Issues: Produces blurred edges and introduces aliasing artifacts at tissue boundaries

B-Splines

Standard: Widely adopted clinical method

Issues: Over-smooths images near sharp intensity transitions, losing fine anatomical detail

General Limitation

Classical interpolation cannot simultaneously preserve fine anatomical structures whilst maintaining smooth intensity variations across slices.

Proposed Hybrid Solution

The "Best of Both Worlds":

A framework that synergistically merges two distinct approaches.

Classical Branch (B-Spline):

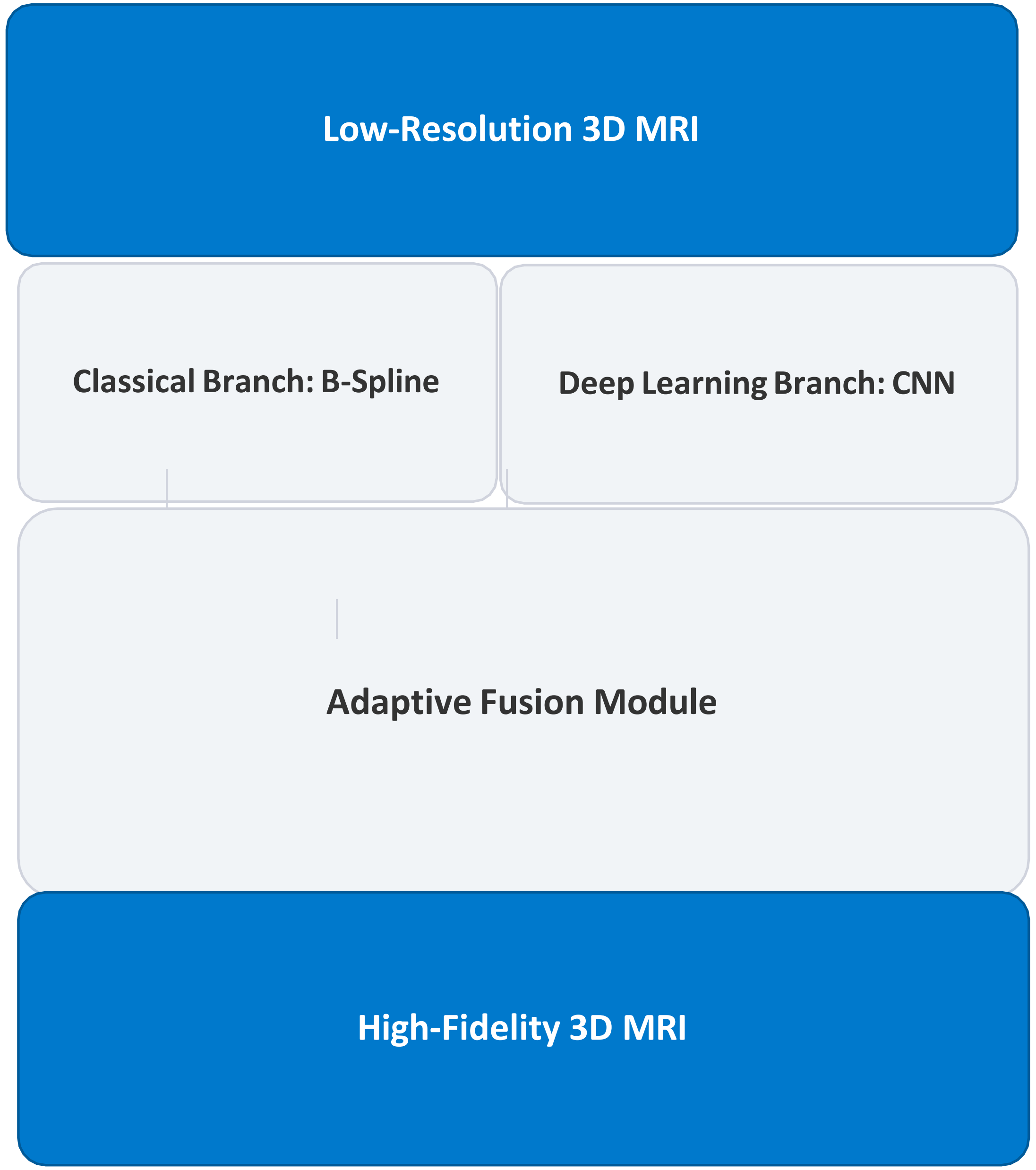
Guarantees numerical stability and smooth structural continuity

Deep Learning Branch (CNN):

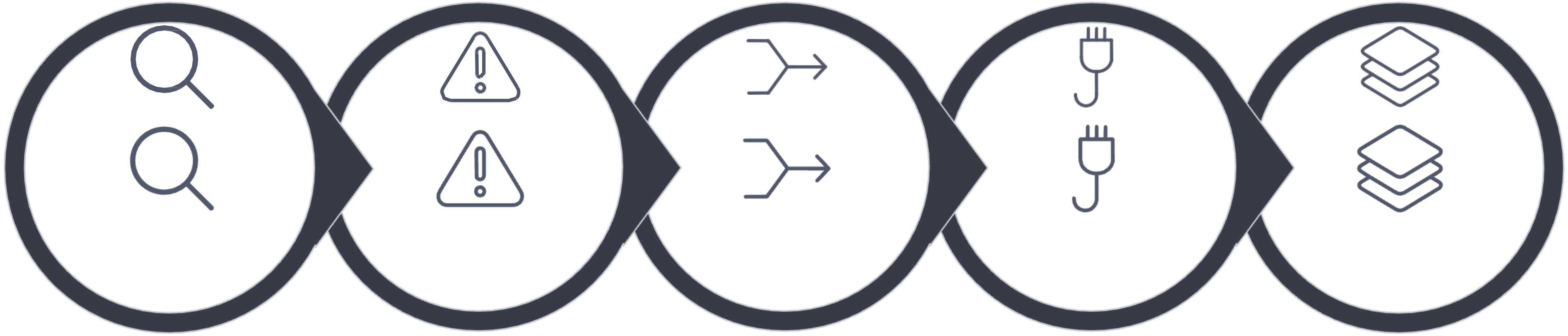
Provides representational power to restore fine textures and sharp edges.

The Result:

Maintains high computational efficiency while significantly improving overall 3D reconstruction fidelity.



Reconstruction Pipeline Architecture



01 **Input Acquisition**
Raw MRI sequence with large slice gaps

02 **Gap Mapping**
Identifying the exact spatial coordinates for slice synthesis

03 **Interpolation Branch**
Classical algorithms generate initial estimates

04 **ML Branch**
CNN/Transformer reconstruct high-frequency anatomical details

05 **Adaptive Fusion**
Weighted merging of both branches

06 **Output Volume**
Reconstructed high-resolution 3D volume

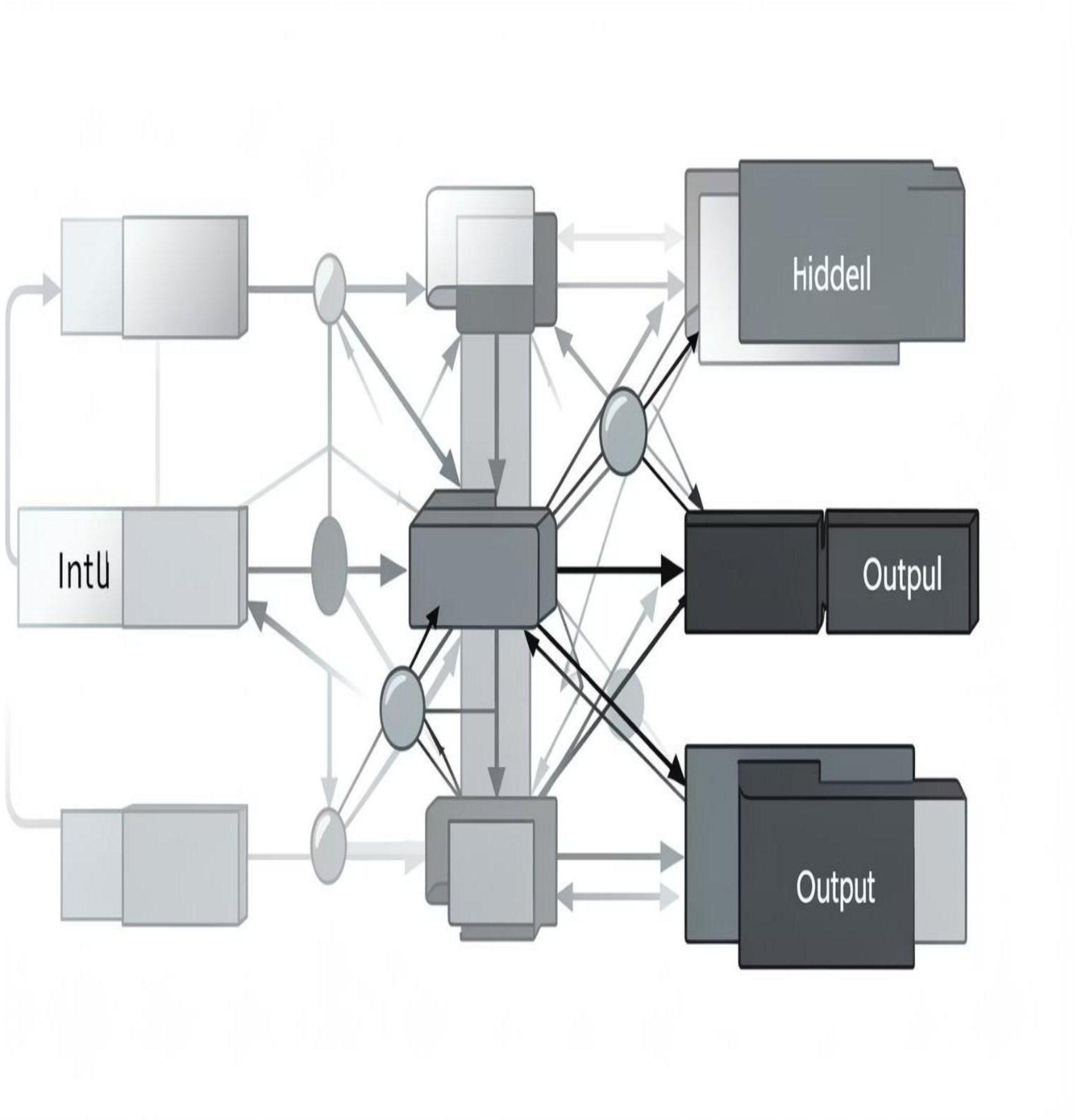
ML Branch Architecture

Structural Prior
Learns the complex patterns of human anatomy from the BraTS dataset.

Residual Learning
The network is trained to predict only the missing high-frequency details (textures and edges).

Spatial Feature Extraction
Utilizes multi-scale convolutional filters to capture hierarchical textures and sharp tissue boundaries.

Efficiency Module
A lightweight design that balances reconstruction fidelity with rapid inference times for clinical environments.



Adaptive Fusion Method

Dynamic Weighting Formula

Final reconstruction combines both branches using learnable weights:

Output = $\alpha \times$ ML + $(1 - \alpha) \times$ Interpolation

$$\text{Output} = w \times \text{ML} + (1 - w) \times \text{Interpolation}$$

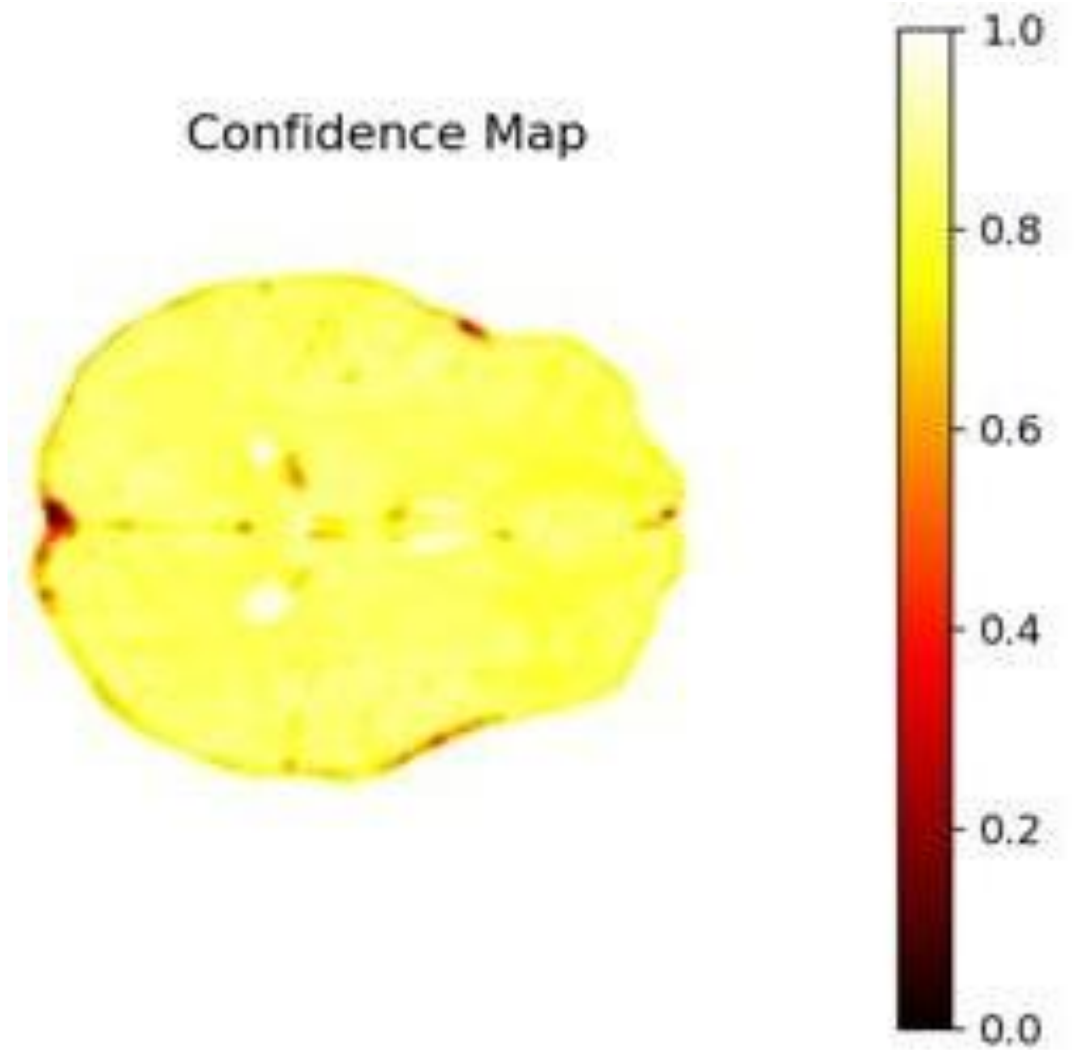
Aware Adaptation

The fusion module identifies high-frequency regions (edges/lesions) and dynamically assigns higher weights to the Deep Learning branch to restore sharp anatomical detail.

Structural Control

Uses the classical branch as a "ruler" to double-check the AI's work. Prevents the system from creating fake textures or artifacts. Ensures the final 3D reconstruction keeps the patient's real physical shape.

Confidence Map Generation



Uncertainty Quantification

The Confidence Score

The system generates a map where every pixel (voxel) is assigned a value from 0 to 1.

Interpretation

High values (yellow/white) show where the reconstruction is certain, while low values (red/black) highlight areas of uncertainty.

Targeted Enhancement

The map identifies exactly which regions relied on Deep Learning for detail versus those that used classical math for stability.

Verification

This allows doctors to "double-check" the AI's work in high-risk areas like tumor boundaries.

Reliability

Increases clinical trust by showing that the model "knows when it doesn't know."

Performance Comparison Analysis

36.4 dB

PSNR Improvement

Higher PSNR indicates better reconstruction quality with reduced noise compared to classical interpolation methods

0.98

SSIM Increase

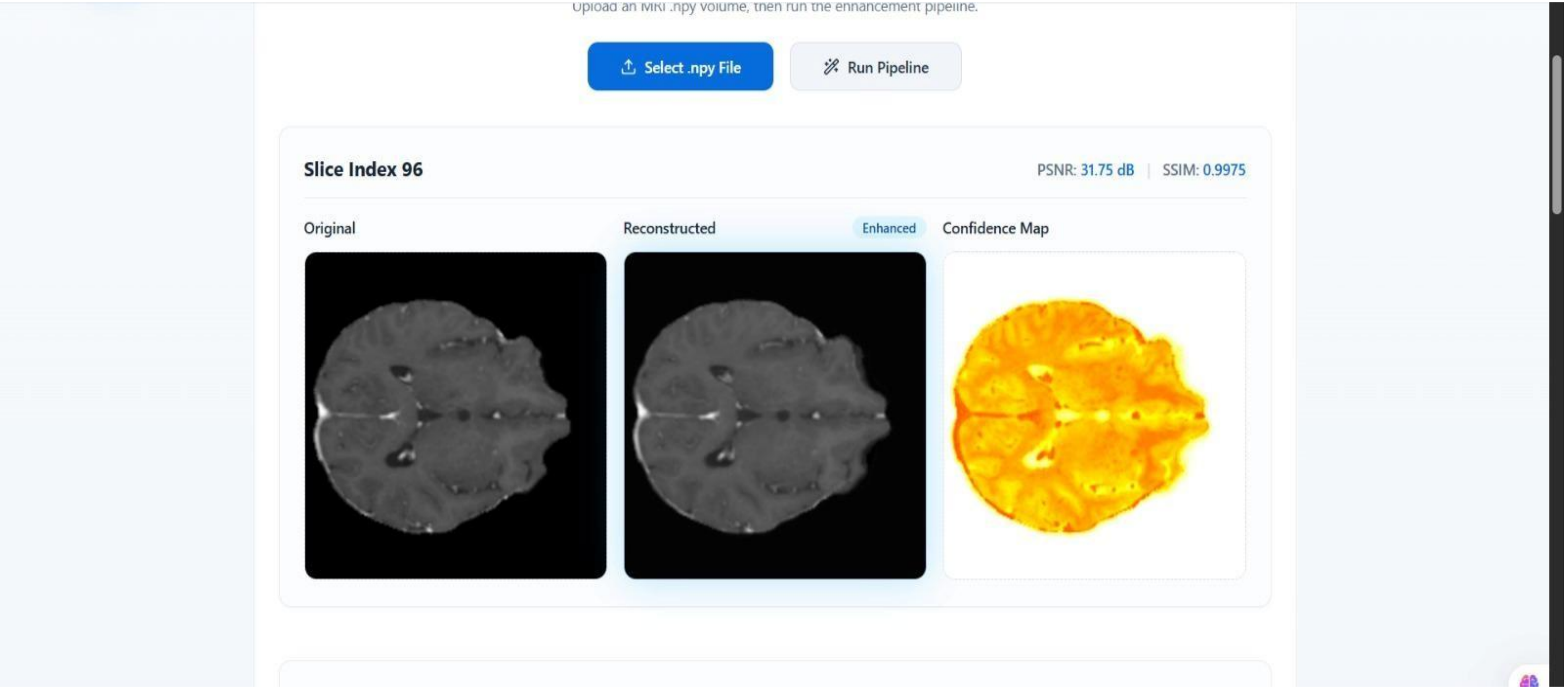
High SSIM value shows strong structural similarity between reconstructed and original MRI slices.

Quantitative results demonstrate that the proposed hybrid method significantly improves image quality compared to traditional interpolation techniques.

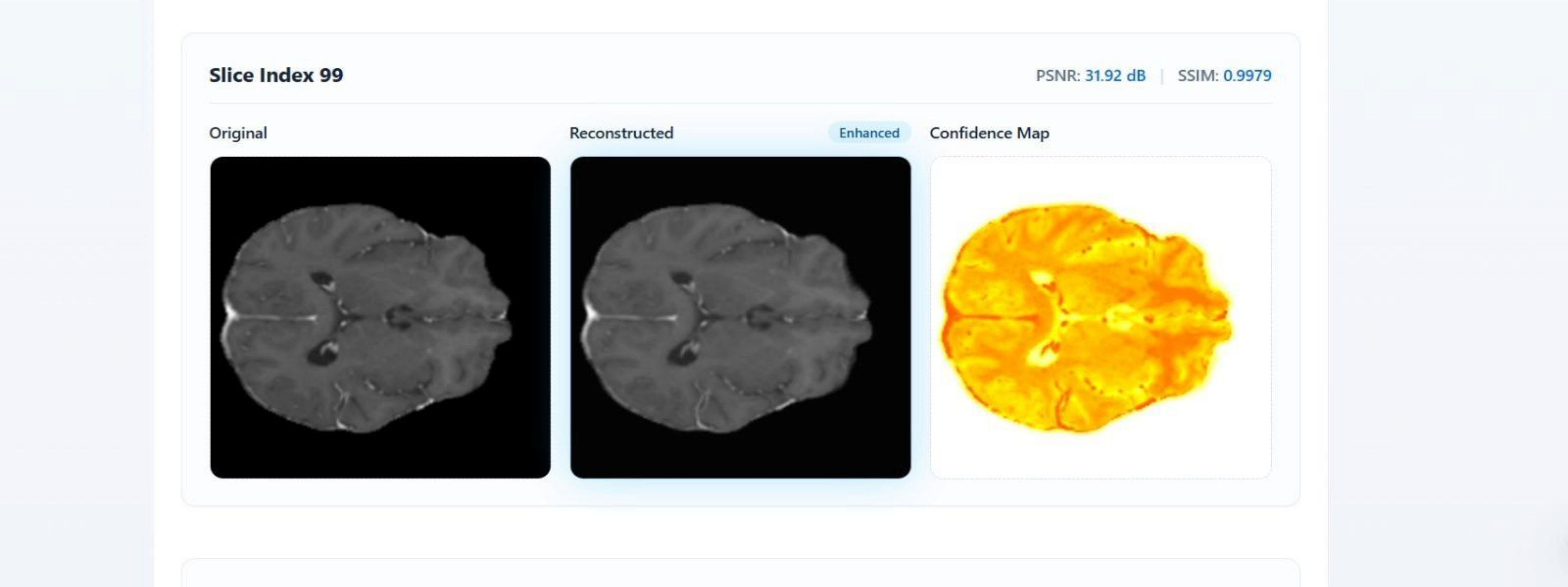
Note: Results are computed on test MRI data and show consistent improvement over baseline methods.

Experimental Results

Experimental Results



Experimental Results



Advantages of the Hybrid System

Superior Image Quality

Preserves fine anatomical details whilst maintaining smooth intensity variations across slices.

Numerical Stability

Classical interpolation branch provides robust foundation, preventing pathological network behaviour.

Computational Efficiency

Hybrid architecture maintains reasonable inference times suitable for clinical deployment.

Clinical Applicability

Confidence maps enable uncertainty-aware interpretation and automated quality assurance.

CONCLUSION

- Successfully enhances low-resolution MRI images into high-resolution outputs using a hybrid approach
- B-spline interpolation ensures smooth structural reconstruction and preserves anatomical continuity
- CNN effectively restores fine details, edges, and textures beyond traditional interpolation
- Hybrid approach → improved accuracy and visual quality
- Achieves superior image quality metrics, including higher PSNR and improved SSIM
- Capable of handling missing or sparsely sampled MRI slices for partial data reconstruction
- Demonstrates strong potential for 3D MRI reconstruction from 2D slices
- Supports improved medical diagnosis through clearer and more reliable images
- Scalable framework, extendable with advanced models such as GANs and attention-based networks
- Confirms that combining classical mathematical techniques with deep learning yields superior MRI enhancement results